



PALM: Physics-Aware Leakage Minimizer

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Motivation

- Machine learning models in drug discovery, materials science and bioinformatics are often evaluated with datasets with data leakage
- Goal: Provide a unified, automated tool for leakage-minimized splitting across diverse scientific data domains.

Training set
Drug A, Drug B, ...

Test set
Drug C, Drug D, ...

If drugs in the training set are similar to drugs in the test set, then there is probability of data leakage

Background

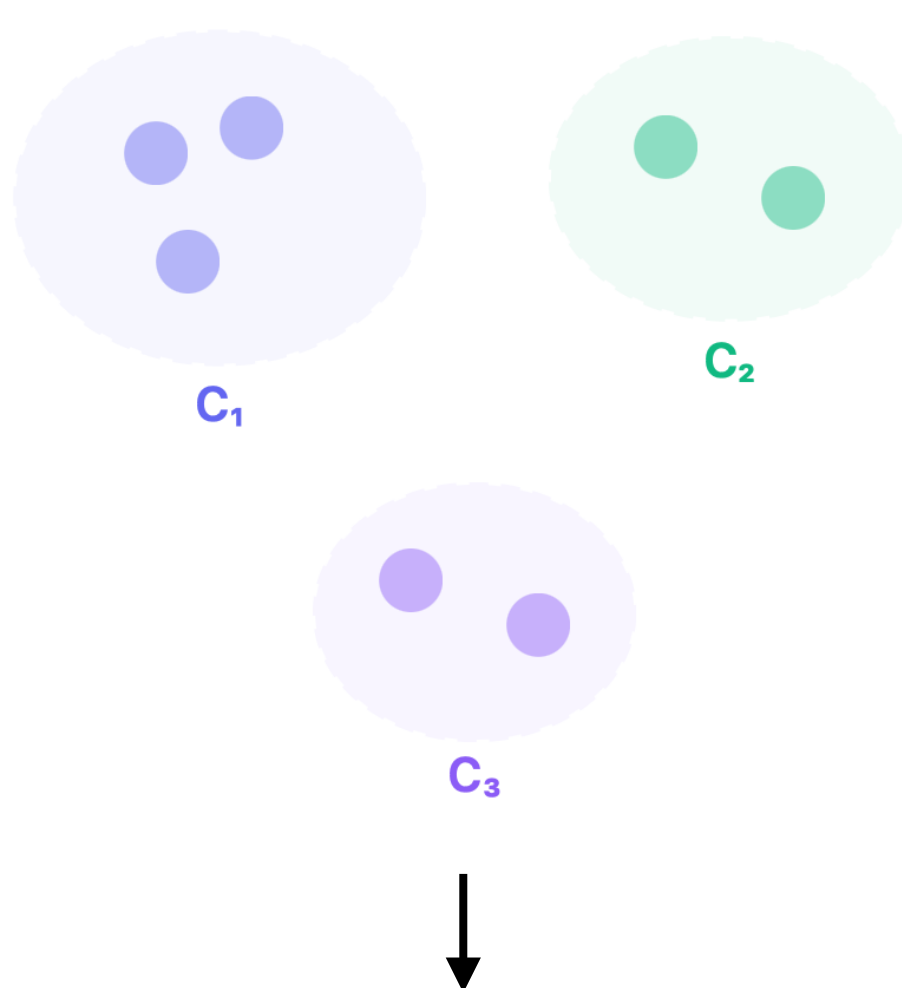
DataSAIL (Data Splitting Against Information Leakage)

formulates splitting as a constrained optimization problem, minimizing cross-split similarity while respecting ratio constraints

Pairwise similarity matrix

X1	1.0	.85	.12	.08	.72	.10
X2	.85	1.0	.09	.05	.78	.07
X3			1.0	.91		
X4				1.0		
X5					1.0	
X6						1.0
X1	X2	X3	X4	X5	X6	

Cluster into groups



Decision Variables
 $\pi(x) \in \{1, \dots, k\}$
Assign each cluster x to fold $\pi(x)$
Integer constraint: fold IDs are discrete

Objective: Minimize Leakage
$$\min \sum \text{sim}(x, x')$$

$$\forall x, x' : \pi(x) \neq \pi(x')$$

↑ cross-fold similarity is minimized

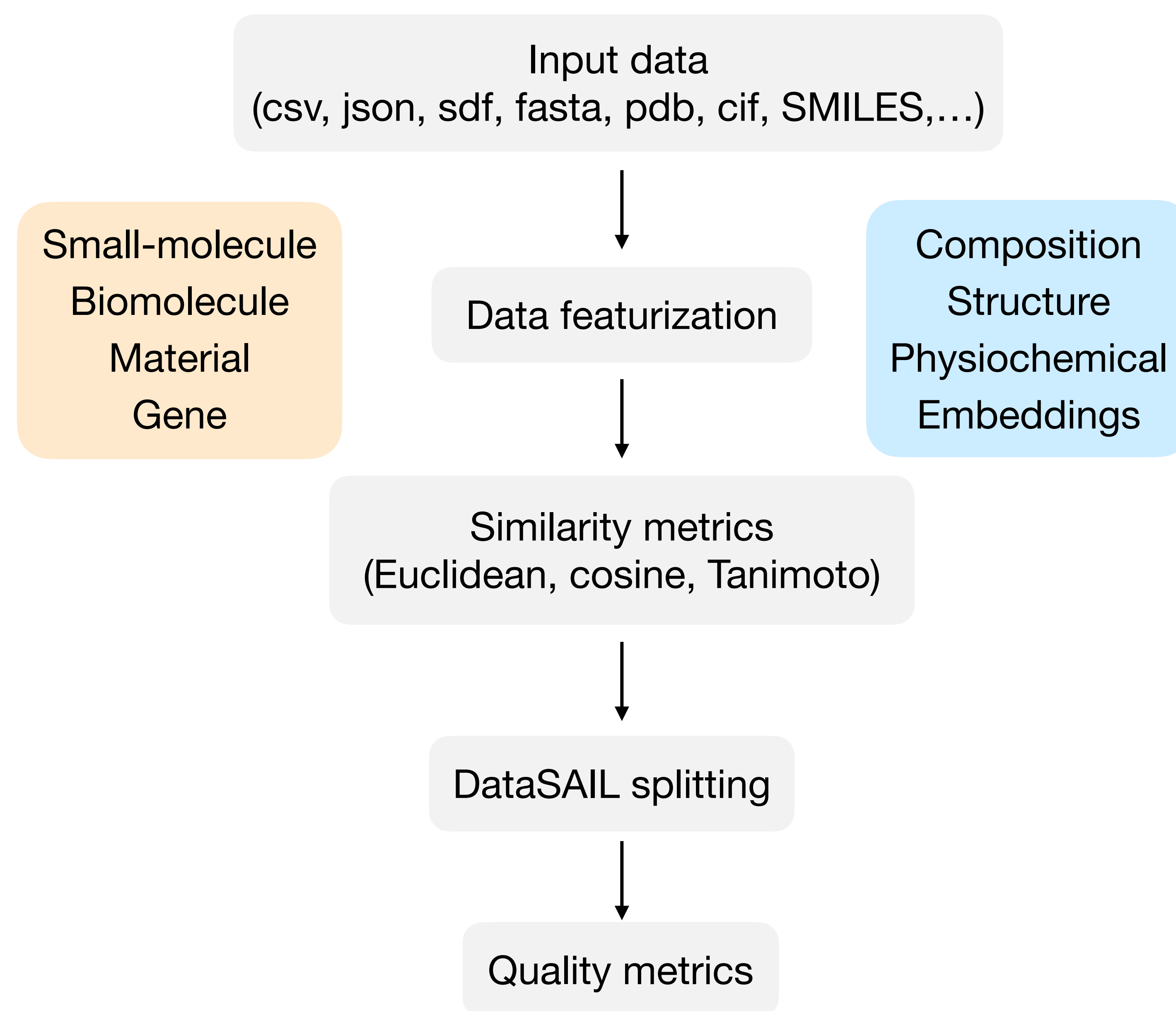
Subject to Constraints
1 **Fold Size:**
 $|\text{fold}_i| \approx s_i \cdot n$ (within tolerance ϵ)
2 **Stratification:**
 $\text{class ratio per fold} \approx \text{global ratio}$ (within δ)

Solvers (via CVXPY)
GUROBI MOSEK SCIP

PALM extends DataSAIL with:

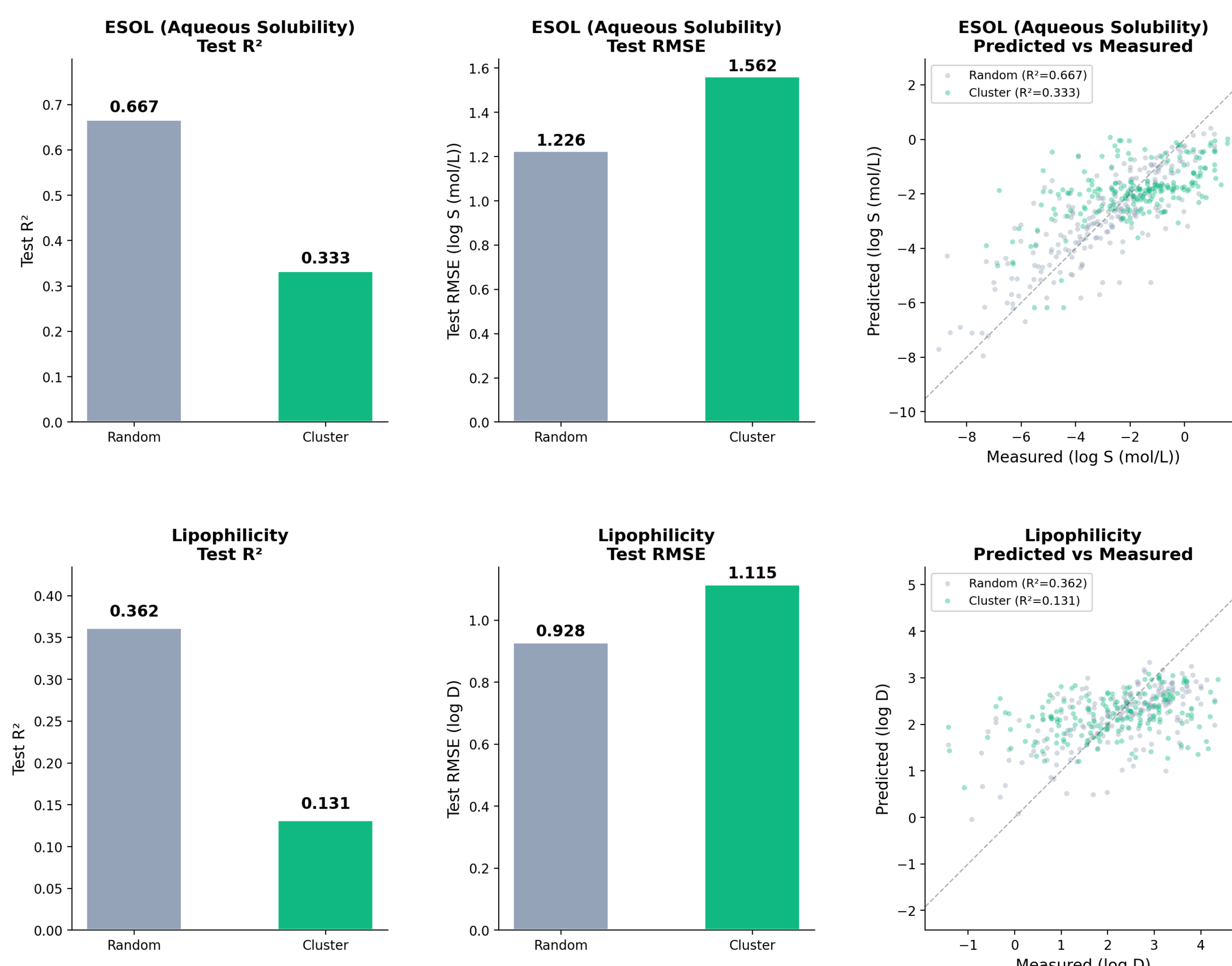
- Domain-specific feature extraction for 4 entity types
- Adaptive distance metrics
- Support for 14 input format with automatic detection
- Quality metrics and visualization
- An interactive web application

Workflow



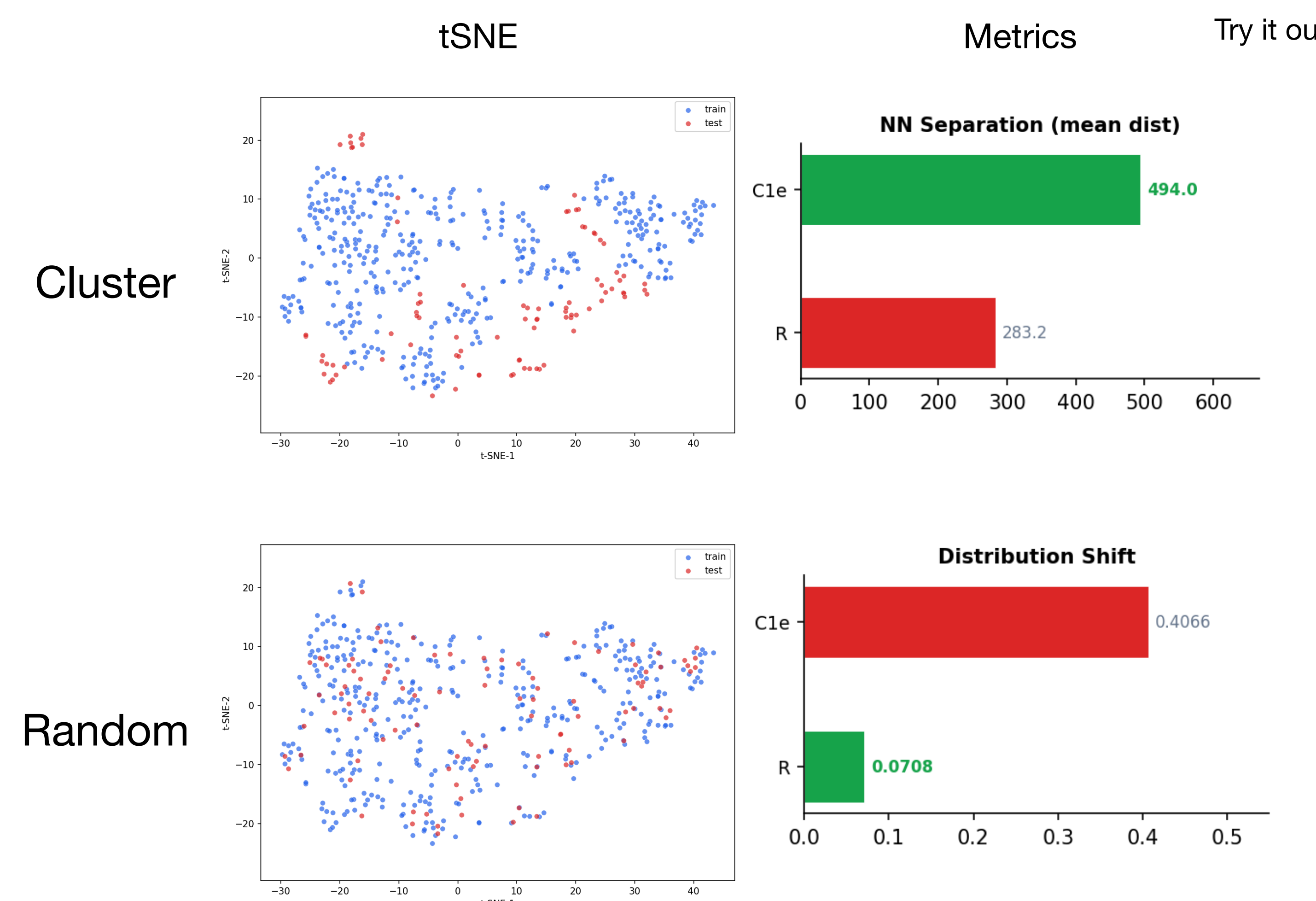
Preliminary Results

Molecular dataset



- Random splitting inflates R^2 by more than 2x on molecular property prediction
- Cluster-based splitting provides realistic performance estimates by eliminating structural similarity leakage

Materials dataset



Discussion

- Unified framework** for leakage-aware splitting across four scientific domains (molecules, biomolecules, materials, genes) with a single configuration interface
- Adaptive algorithm selection:** automatic distance metric and dimensionality reduction based on feature sparsity
- Format preservation:** input and output in the same format, enabling direct integration into existing ML workflows
- Quantitative evaluation:** built-in metrics (nearest neighbor leakage, distribution shift, coverage) provide objective evidence of split quality
- Web interface for non-programmers:** The GUI with live progress, automatic column detection, and downloadable results lowers the barrier for experimentalists who don't use command-line tools

References

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